

1 Solution — GIAM 6.3

Problem 3

1.1 Problem

Define a relation A on the set of all words by

$$w_1 A w_2 \iff w_1 \text{ is an anagram of } w_2.$$

Show that A is an equivalence relation. (Words are anagrams if the letters of one can be re-arranged to form the other; e.g. “ART” and “RAT”.)

1.2 Setup

Let Σ be the alphabet — a fixed finite set of letters — and let W be the set of all finite **words** over Σ , i.e. finite sequences of letters from Σ . (“Word” here means any finite string of letters; dictionary status plays no role in the relation’s definition.)

To formalize “rearrangement,” send each word to its multiset of letters. For $w \in W$, let

$$\mu(w) = \text{the multiset of letters appearing in } w, \text{ counted with multiplicity.}$$

Equivalently, $\mu(w) : \Sigma \rightarrow \mathbb{Z}_{\geq 0}$ sends each letter to the number of times it occurs in w , so $\mu : W \rightarrow M(\Sigma)$, where $M(\Sigma)$ denotes the set of finite multisets over Σ .

The definition of “anagram” is exactly:

$$w_1 A w_2 \iff \mu(w_1) = \mu(w_2).$$

For instance, $\mu(\mathbf{ART}) = \mu(\mathbf{RAT}) = \text{”one A, one R, one T,”}$ so $\mathbf{ART} A \mathbf{RAT}$.

This is again “equality pulled back through a function” (cf. 6.2 #2 and 6.3 #2), so the three equivalence properties follow from the corresponding properties of equality on $M(\Sigma)$.

1.3 A Is an Equivalence Relation

- **Reflexive.** $\mu(w) = \mu(w)$, so wAw . Every word is trivially a rearrangement of itself.
- **Symmetric.** If w_1Aw_2 , then $\mu(w_1) = \mu(w_2)$; equality is symmetric, so $\mu(w_2) = \mu(w_1)$, hence w_2Aw_1 . (If a permutation rearranges w_1 into w_2 , the inverse permutation rearranges w_2 back to w_1 .)
- **Transitive.** If w_1Aw_2 and w_2Aw_3 , then $\mu(w_1) = \mu(w_2)$ and $\mu(w_2) = \mu(w_3)$; equality is transitive, so $\mu(w_1) = \mu(w_3)$, hence w_1Aw_3 . (Composing two rearrangements is itself a rearrangement.)

All three hold, so A is an equivalence relation. ■

1.4 Equivalence Classes

For each $w \in W$,

$$[w]_A = \{v \in W : \mu(v) = \mu(w)\},$$

the set of all words sharing w 's letter-multiset. For example,

$$[\mathbf{ART}]_A = \{\mathbf{ART}, \mathbf{ATR}, \mathbf{RAT}, \mathbf{RTA}, \mathbf{TAR}, \mathbf{TRA}\}$$

— the $3! = 6$ permutations of three distinct letters. When letters repeat, the count drops; for instance $[\mathbf{LOOK}]_A$ has size $4!/(1! \cdot 1! \cdot 2!) = 12$, since the two O's are interchangeable.

In general, if w has length n and its distinct letters appear with multiplicities $k_1, \dots, k_m$ (so $k_1 + \dots + k_m = n$), then

$$|[w]_A| = \frac{n!}{k_1! k_2! \dots k_m!},$$

the multinomial coefficient.

1.5 Remark — The Quotient Set

The map $[w]_A \mapsto \mu(w)$ is a bijection from W/A onto $M(\Sigma)$: each anagram class corresponds to exactly one multiset of letters, and every finite multiset of letters arises as $\mu(w)$ for some word (any arrangement of its elements). So

$$W/A \longleftrightarrow M(\Sigma),$$

and “anagram class” is simply another name for “multiset of letters.”